

Week # 2.

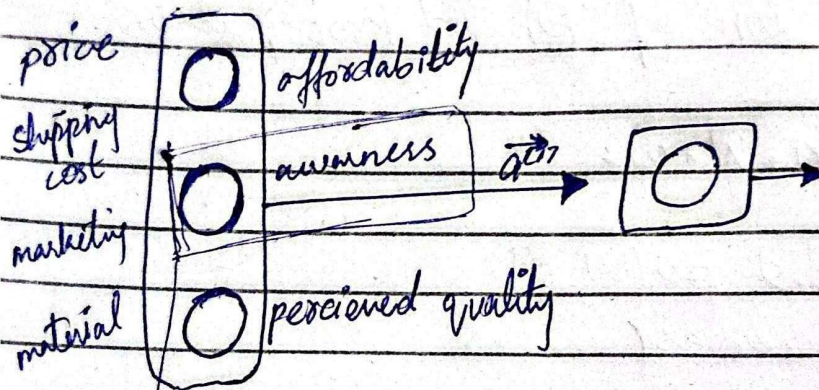
Training of a neural network.

- ① Defining the model
- ② Loss and cost function
- ③ Gradient Descent

↳ "Back propagation"



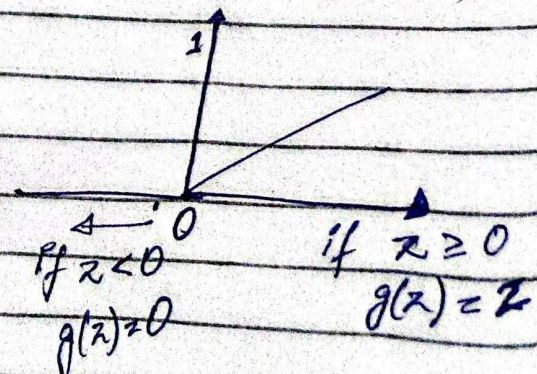
* Activation Functions



Awareness was supposed to be 0 or 1 but users can aware of a product or a little aware or even it was viral so awareness $\gg 1$

So a better activation function would be ~~softmax~~ ReLU which is $g(z) = \max(0, z)$

Rectified linear unit



SOFTMAX

How to choose Activation Functions for neurons?

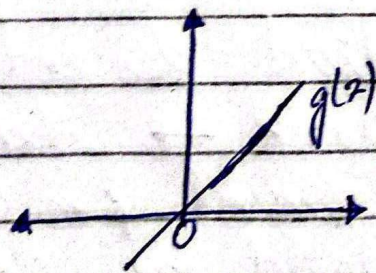
Output Layer:-

The activation function for output layer is more dependent to the ground truth if we want binary classification then sigmoid is a natural choice.

Sigmoid for Binary classification

Regression problems like stock prices
 Y (ground truth) = $\pm$ number.
then use Linear Regression function

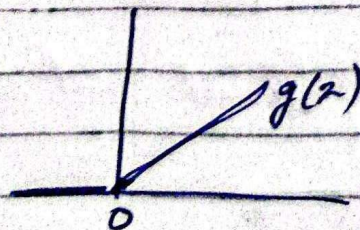
e.g.
stock
prices



x_1, x_2	Y
-1.5	+1
-9	+180

Regression where $Y = +$ numbers only
then ReLU is a good choice
 $Y = 0$ or $+$ ve

e.g., house
prediction



x_1, x_2	Y
150	
250	
300	

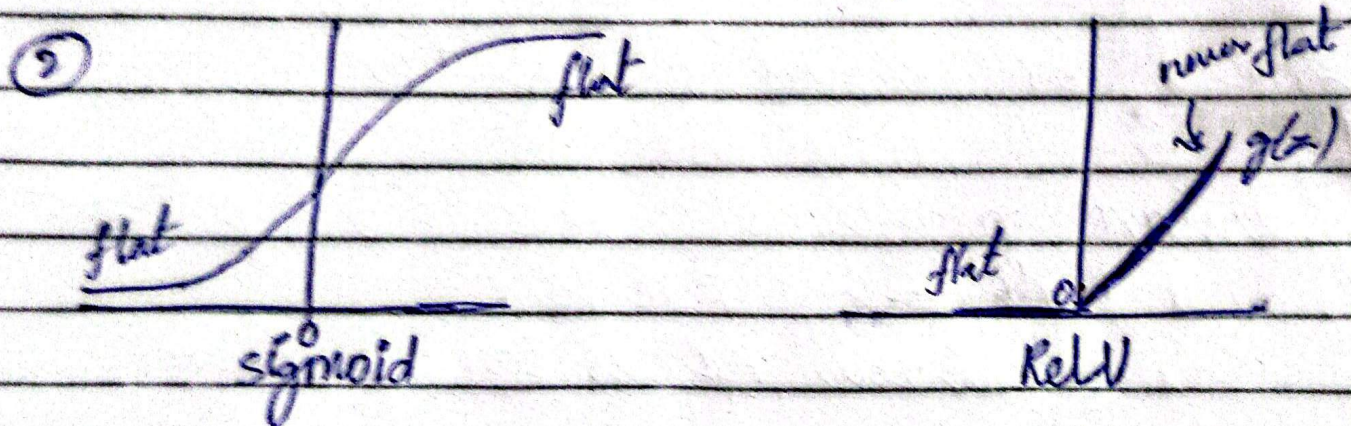
→ HIDDEN LAYERS

By far ReLU is most common activation function for hidden layers.

Why not sigmoid and why ReLU?

① Computation overhead in Sigmoid is large $g(z) = \frac{1}{1+e^{-z}}$ so slower.

while ReLU is literally just (0 or z)



When using gradient descent to train Neural Network → when function which is flat over several places gradient descent would be slow.
local minimum problem.

